

# Inflation Without an Inflaton: Observer-Screen Synchronization in Observer-Patch Holography Cosmology

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## Abstract

Observer-Patch Holography (OPH) cosmology replaces the inflaton with finite observer-screen synchronization. Flatness, horizon coherence, near scale invariance, hot source release, and acoustic-transfer inputs are treated as jobs for the finite screen state rather than for a new scalar field.

The conditional source theorem fixes the screen amplitude from a primitive collar release law and the tilt from an edge-center generator. The radial result proves one-shell non-identifiability and supplies physical dilation and cross-covariance tomography as conditional uniqueness routes. A physical cosmological prediction requires one finite source instantiation, transfer calculation, and likelihood; simulator outputs without those receipts are diagnostics.

## 1 Scope

This technical companion states the inflation-free branch against the compact, screen-microphysics, dark-sector, and finite-source cosmic microwave background (CMB) papers.

## 2 Finite quotient theorem scope

**Finite quotient ensemble theorem.** Fix a finite regulator  $r$ . The physical presentation space is  $\Sigma_r$ , the presentation redundancy groupoid is  $\Gamma_r$ , and the finite physical quotient is

$$Q_r = \Sigma_r / \Gamma_r, \quad \pi_r : \Sigma_r \rightarrow Q_r.$$

The quotient removes nonphysical presentation data: gauge representatives, port relabelings, mesh labels, shard or worker identifiers, queue order, repair schedule identifiers, retry counters, timestamps unless declared semantic, hidden carrier coordinates, and inert ancillary labels. If only settled configurations carry probability, the probability space is the normal-form subset

$$N_r = n_r(Q_r).$$

The map  $n_r$  is a normal-form map, not a probability law. Any physical branch inherits this firewall: it must declare the quotient-intrinsic source law or action before a normal form can be read as a selection or prediction claim.

**Observable algebras and reference states.** In the finite classical case the quotient observable algebra is

$$\mathcal{O}_r = \ell^\infty(Q_r).$$

In the finite quantum case the physical algebra is a declared quotient algebra  $\mathcal{A}_r^{\text{phys}}$  with state

$$\omega_r(A) = \text{Tr}(\rho_r A).$$

When the reference object is obtained from a finite lifted carrier, the load-bearing data are not an abstract groupoid cardinality alone. They are a tracially pointed quotient

$$\left(\mathcal{A}_{r,b}^{\text{phys}}, \tau_{r,b}^0\right), \quad \mathcal{A}_{r,b}^{\text{phys}} = z_{r,b} B(\tilde{\mathcal{H}}_r)^{G_r} z_{r,b},$$

where  $U_r : G_r \rightarrow U(\tilde{\mathcal{H}}_r)$  is the compact gauge action and  $z_{r,b}$  is the central projection for the declared boundary or superselection sector. The reference trace is

$$\tau_{r,b}^0(A) = \frac{\text{Tr}_{\tilde{\mathcal{H}}_r}(A)}{\text{Tr}_{\tilde{\mathcal{H}}_r}(z_{r,b})}.$$

If

$$z_{r,b} \tilde{\mathcal{H}}_r \cong \bigoplus_{\alpha} V_{\alpha} \otimes M_{\alpha},$$

with  $d_{\alpha} = \dim V_{\alpha}$  and  $m_{\alpha} = \dim M_{\alpha}$ , then

$$\mathcal{A}_{r,b}^{\text{phys}} \cong \bigoplus_{\alpha} I_{V_{\alpha}} \otimes B(M_{\alpha}), \quad p_{r,\alpha} = \frac{d_{\alpha} m_{\alpha}}{\sum_{\beta} d_{\beta} m_{\beta}}.$$

These are the induced central-sector weights only after the carrier representation and boundary sector have been fixed.

**OPH quotient ensemble.** An OPH quotient ensemble is specified by a quotient-intrinsic base weight and action

$$m_r : Q_r \rightarrow \mathbb{R}_{>0}, \quad S_r : Q_r \rightarrow \mathbb{R} \cup \{+\infty\},$$

and

$$w_r(q) = m_r(q) e^{-S_r(q)}, \quad Z_r = \sum_{q \in Q_r} w_r(q), \quad \mu_r(q) = Z_r^{-1} w_r(q).$$

Equivalently, one may state an intrinsic projective prior  $\nu_r$  on  $Q_r$  and set  $\mu_r = (n_r)_{\#} \nu_r$ . Uniform quotient counting, uniform representative counting pushed to the quotient, groupoid weights, and tracial central-sector weights are different physical claims. The paper must declare which one is being used.

**Normal-form projector non-selection.** For any retraction  $N : Q \rightarrow Q_{\text{nf}}$  onto a subset  $Q_{\text{nf}} \subseteq Q$ , that is, any map whose restriction to  $Q_{\text{nf}}$  is the identity, the induced map on laws

$$\mathcal{C}_Q(\mu) = N_{\#} \mu$$

is idempotent:

$$\mathcal{C}_Q^2 = \mathcal{C}_Q.$$

Every law supported on  $Q_{\text{nf}}$  is fixed; both statements use the retraction property. Therefore settlement or canonicalization never selects a unique physical probability law by itself.

**Selection-gap corollary.** Let  $X \subseteq Q_{\text{nf}}$  be a finite set of quotient-normal candidates distinguished by visible invariants. Normal-form data determine  $X$  and its quotient-visible invariants, but they do not choose a member of  $X$ . If two laws  $\mu, \nu$  are supported on  $X$  and concentrate on different candidates, both are fixed by  $\mathcal{C}_Q$ . Unique sector selection therefore requires source data: an intrinsic action with a unique minimizer, a declared physical ensemble, or a refinement-stable gap certificate. A defect or holonomy classification can classify possible sectors without choosing the physical sector, and a contraction or repair generator can certify convergence toward a declared target without creating the target law.

**Finite MaxEnt quotient ensemble.** For finite  $Q$ , positive  $m$ , and quotient observables  $F_1, \dots, F_k$ , maximizing

$$\mathcal{H}_m(\nu) = - \sum_q \nu(q) \log \frac{\nu(q)}{m(q)}$$

subject to

$$\sum_q \nu(q) = 1, \quad \sum_q \nu(q) F_a(q) = c_a$$

has the full-support solution, when the feasible full-support surface is nonempty,

$$\mu(q) = \frac{m(q) \exp[-\sum_a \theta_a F_a(q)]}{Z(\theta)}.$$

Boundary optima obey the same formula after restricting to their support. On a finite noncommutative quotient algebra with faithful reference state  $\sigma_r$ ,

$$\rho_r = \frac{\exp(\log \sigma_r - \sum_a \theta_a F_{r,a})}{\text{Tr} \exp(\log \sigma_r - \sum_a \theta_a F_{r,a})}.$$

The finite constraint ledger must name every  $F_{r,a}$ , its units and support, the target expectation and source, sector or zero-mode treatment, refinement transformation, and proof that no run output or observational output entered the source definition.

**Refinement compatibility and RG closure.** For  $s \succeq r$ , let  $c_{sr} : Q_s \rightarrow Q_r$  be the physical coarse map. Exact compatibility of weighted ensembles is equivalent to the fiber-sum identity

$$\sum_{q': c_{sr}(q')=q} m_s(q') e^{-S_s(q')} = \alpha_{sr} m_r(q) e^{-S_r(q)}$$

for a constant  $\alpha_{sr} > 0$  independent of  $q$ . Then

$$(c_{sr})_{\#} \mu_s = \mu_r.$$

If the one-step defects are

$$\delta_{k+1,k} = \|(c_{k+1,k})_{\#} \mu_{k+1} - \mu_k\|_{\text{TV}},$$

then

$$\|(c_{nr})_{\#} \mu_n - \mu_r\|_{\text{TV}} \leq \sum_{k=r}^{n-1} \delta_{k+1,k}.$$

For exponential-family refinement, exact closure requires the fine conditional free energy

$$G_{sr,\theta}(q_r) = -\log \mathbb{E}_{m_s^0} [\exp[-\theta \cdot F_s(Q_s)] \mid c_{sr}(Q_s) = q_r]$$

to equal  $\kappa_{sr}(\theta) + R_{sr}(\theta) \cdot F_r(q_r)$ . If the residual is uniformly bounded by  $\varepsilon$ , the induced total-variation defect is bounded by  $\tanh \varepsilon$ .

**Implementation invariance and representative lifting.** If implementations  $A, B$  have quotient bijections  $h_r : Q_r^A \rightarrow Q_r^B$  satisfying

$$m_r^B(h_r q) = m_r^A(q), \quad S_r^B(h_r q) = S_r^A(q), \quad h_r \circ c_{sr}^A = c_{sr}^B \circ h_s,$$

then

$$(h_r)_\# \mu_r^A = \mu_r^B.$$

For tracially pointed quantum quotients the corresponding equivalence is a trace-preserving quotient equivalence. It is invariant under unitary intertwiners preserving the gauge action and sector, and under inert trivial ancillas  $A \mapsto A \otimes I_{\text{anc}}$ . It is not invariant under arbitrary changes of gauge-representation multiplicities.

If an implementation stores representatives, a representative-level law must be a conditional lift

$$\tilde{\mu}_r(x) = \mu_r(\pi_r x) \kappa_r(x \mid \pi_r x), \quad \sum_{x: \pi_r(x)=q} \kappa_r(x \mid q) = 1.$$

Then  $(\pi_r)_\# \tilde{\mu}_r = \mu_r$ . Uniform representative sampling yields orbit-size weights and is physical only if representative counting is the declared base measure.

**Quotient-lumpable kernels and sampler correctness.** A representative kernel  $\tilde{P}(x, y)$  descends to  $Q_r$  only when

$$P_Q(q, q') = \sum_{y: \pi(y)=q'} \tilde{P}(x, y)$$

is independent of the chosen representative  $x \in \pi^{-1}(q)$ . For  $w(q) = m(q)e^{-S(q)}$  and proposal  $R(q, q')$  with reciprocal support, the Metropolis–Hastings acceptance rule

$$a(q, q') = \min \left\{ 1, \frac{w(q')R(q', q)}{w(q)R(q, q')} \right\}$$

gives detailed balance

$$\mu(q)R(q, q')a(q, q') = \mu(q')R(q', q)a(q', q).$$

Repair-informed proposals must include the Hastings asymmetry term; otherwise the stationary law is generically changed.

**Repair generators are not selectors.** A repair generator of the form

$$L_{\text{rep}} = \sum_C c_C (I - E_C)$$

is a relaxation or sampling object after a law has been selected. Conditional expectations  $E_C$  are defined on  $L^2(X_r, \pi_r)$ , so the reference law  $\pi_r$  is input. On overlapping collars the expectations need not commute. The correct finite gap certificate is the Poincare constant

$$\kappa_r = \inf_{f \perp 1} \frac{\sum_C \|(I - E_C)f\|^2}{\|f\|^2}.$$

If local fiber rates have a positive lower bound  $\gamma_*$ , then

$$L_{\text{rep}} \geq \gamma_* \kappa_r (I - P_0).$$

Finite repair completeness gives  $\kappa_r > 0$  at fixed regulator. A uniform refinement lower bound  $\inf_r \kappa_r > 0$  is a separate theorem or receipt.

**Finite evidence accuracy.** For bounded coarse observables  $O$ , if

$$\|\hat{\mu}_s - \mu_s\|_{\text{TV}} \leq \epsilon_{\text{samp}}$$

and the refinement defects sum to  $\epsilon_{\text{ref}}$ , then

$$\left| \mathbb{E}_{\hat{\mu}_s}[O \circ c_{sr}] - \mathbb{E}_{\mu_r}[O] \right| \leq 2\|O\|_{\infty}(\epsilon_{\text{samp}} + \epsilon_{\text{ref}}).$$

Continuum-facing observables require a realization map and correlation Cauchy bound in addition to a finite histogram.

**Physical-vacuum requirement.** A stationary sampler is not a physical vacuum. For any faithful target law one can build a positive transfer operator with that law as ground state, so positivity alone is not a selector. A physical-vacuum interpretation requires source Euclidean slab data

$$\mathfrak{S}_r^E = (Q_r, m_r^0, J_r, V_r, a_{t,r})$$

whose conductance  $J_r(q, q') = J_r(q', q) \geq 0$ , local potential  $V_r$ , and slab thickness  $a_{t,r}$  are derived without using the target law or sampler output. With connected event graph,

$$(H_r^E f)(q) = \frac{1}{m_r^0(q)} \sum_{q'} J_r(q, q') (f(q) - f(q')) + V_r(q) f(q)$$

is self-adjoint and bounded below on  $L^2(Q_r, m_r^0)$ ; its finite Feynman–Kac semigroup is positivity improving. Perron–Frobenius gives a unique positive normalized ground state  $\Omega_r$ , and the finite vacuum law is

$$\mu_r^{\text{vac}}(q) = |\Omega_r(q)|^2 m_r^0(q).$$

For  $T_r = e^{-a_{t,r}(H_r^E - E_{0,r})}$ , the Doob kernel is stochastic and detailed-balanced with  $\mu_r^{\text{vac}}$ . A continuum interpretation additionally requires reflection positivity or equivalent reconstruction plus refinement compatibility of the transfer family.

**Primordial and cosmological prediction firewall.** A screen covariance contains incomplete radial information. The complete one-shell map

$$C_\ell = 4\pi \int_0^\infty \frac{dk}{k} \Delta_\zeta^2(k) j_\ell^2(k\chi_\star)$$

has an infinite-dimensional kernel that persists under positivity. OPH physical primordial interpretation requires the source-only stress, single-clock, entropy-repair, curvature-evolution, adiabatic-mode, phase-coherence, physical mode, radial-null-space, and forward-projection receipts together with a scale-natural physical dilation intertwiner or complete radial cross-covariance tomography. A finite radial prior produces a conditional continuation. Observable CMB comparison also requires declared source, solver, dataset, covariance, nuisance, data-use, and pooled-reducer provenance.

**Interpretation classes and required receipts.** Every ensemble-facing run records its ensemble id, interpretation class, regulator, representative schema, gauge action, canonicalizer, base measure, action coefficients, coarse maps, zero-mode projector, amplitude convention, sampler, smoothing

policy, source provenance, and explicit nonclaims. The seed belongs to the run receipt rather than the ensemble definition. The interpretation classes are

seed, proposal, or repair noise,  
 conventional reference ensemble,  
 OPH-native quotient ensemble,  
 OPH physical vacuum,  
 OPH primordial field,  
 observable cosmological prediction.

The evidence bundle must keep separate receipts for stationary-law schedule invariance, detailed balance of the aggregate kernel, and pathwise partition invariance. Deterministic replay of semantic random streams or a canonical serial chain is useful, but it is not pathwise partition invariance. Smoothing must preserve raw coefficients, raw spectra, smoothing kernels, smoothed coefficients, smoothed spectra, and hashes of each stage; it is not part of  $S_r$  unless explicitly declared.

### 3 Finite Screen Spectrum Theorem Package

The screen-level spectrum theorem supplies the finite source object used by the cosmology branch. It does not supply temperature-temperature, temperature-polarization, polarization-polarization, lensing, likelihoods, or a physical cosmic microwave background interpretation. Those require the Boltzmann and data-contract theorems.

**Definition 3.1** (finite screen regulator). *For regulator  $r$ , a finite screen regulator is*

$$\mathfrak{S}_r = (\mathcal{T}_r, M_r, \Gamma_r, \mathcal{Q}_r, C_{sr}, J_{rs}),$$

where  $\mathcal{T}_r$  is a finite cellulation of the screen,  $M_r \succ 0$  is the area/quadrature mass matrix,  $\Gamma_r$  is the hidden-presentation groupoid,  $\mathcal{Q}_r = \Sigma_r/\Gamma_r$  is the physical quotient, and  $C_{sr}$ ,  $J_{rs}$  are the coarse-graining and interpolation maps. Shape regularity and quadrature convergence require

$$f^\top M_r g \rightarrow \int_{S^2} f g d\Omega$$

with an explicit band-limited error bound

$$\left| f^\top M_r g - \int f g d\Omega \right| \leq \varepsilon_{M,r}(L) \|f\|_{H^s} \|g\|_{H^s}.$$

Patch count alone is therefore not an angular-resolution certificate.

**Definition 3.2** (geometric screen scalar). *Let the quotient-visible collar-volume readout be*

$$J_{X,r}(x) = \lambda_r(x) \sqrt{\det \sigma_{AB,r}(x)} > 0,$$

and let  $\bar{J}_{X,r}$  be emitted by an independently defined homogeneous or frozen-background branch, not by a CMB fit. Define

$$q_{0,r} = \frac{1}{3} \log \frac{J_{X,r}}{\bar{J}_{X,r}}.$$

On a certified spherical chart, with  $B_r = [1, n_x, n_y, n_z]$ ,

$$\Pi_{<2,r} = B_r (B_r^\top M_r B_r)^{-1} B_r^\top M_r, \quad q_r = (I - \Pi_{<2,r}) q_{0,r}.$$

On an irregular screen,  $B_r$  is replaced by the certified generalized eigenprojector onto the finite  $\ell = 0, 1$  subspace. Hard-coded feature  $z$ -scores or observer-summary weights are diagnostic proxies, not  $q_r$ .

**Proposition 3.3** (quotient invariance). *If a recharting  $U$  preserves the screen inner product and geometric readout,*

$$U^\top M'_r U = M_r, \quad q'_{0,r} = U q_{0,r}, \quad B'_r = U B_r,$$

*then  $\Pi'_{\geq 2,r} U = U \Pi_{\geq 2,r}$ , and  $q'_r = U q_r$ .*

*Proof.* Substitution gives

$$\begin{aligned} \Pi'_{< 2,r} U &= U B_r (B_r^\top U^\top M'_r U B_r)^{-1} B_r^\top U^\top M'_r U \\ &= U B_r (B_r^\top M_r B_r)^{-1} B_r^\top M_r = U \Pi_{< 2,r}. \end{aligned}$$

Subtracting from  $U$  gives the high-mode identity.  $\square$

**Definition 3.4** (physical scalar precision and action). *The scalar precision  $K_r$  must have a physical origin on  $V_r = \text{im } \Pi_{\geq 2,r}$ . Allowed branches are:*

$$K_r^{\text{Hess}} = \nabla^2 \Phi_r \Big|_{q=0},$$

*for a quotient-visible mismatch or release free energy; a repair Dirichlet form*

$$K_r^{\text{diss}} = H_r - T_r^\top H_r T_r, \quad T_r^\top H_r T_r \preceq H_r;$$

*or the reversible generator form*

$$K_r^{\text{Dir}} = -\frac{1}{2}(L_r + L_r^\dagger).$$

*A raw nonsymmetric repair matrix or caller-supplied  $\kappa$  is not a precision operator. The absolute normalization must be fixed independently, for example by*

$$K_0 Y_{\ell m} = \ell(\ell + 1) Y_{\ell m}$$

*on the  $\theta = 0$  branch. Once normalized,*

$$S_{\text{scr},r}[q] = \frac{1}{2A_{q,r}} \langle q, K_r q \rangle_r = \frac{1}{2A_{q,r}} q^\top M_r K_r q.$$

**Theorem 3.5** (finite MaxEnt screen covariance). *Let  $K_r e_i = \kappa_i e_i$  with an  $M_r$ -orthonormal basis of  $V_r$ ,  $\kappa_i > 0$ , and  $d_r = \dim V_r$ . Among continuous densities in coefficients  $q = \sum_i a_i e_i$  with  $\mathbb{E}[a_i] = 0$  and*

$$\frac{1}{2} \mathbb{E} \langle q, K_r q \rangle_r = E_{q,r}^{\text{src}},$$

*the unique entropy maximizer is*

$$d\mu_r(q) = Z_r^{-1} \exp \left[ -\frac{1}{2A_{q,r}} \langle q, K_r q \rangle_r \right] dq, \quad A_{q,r} = \frac{2E_{q,r}^{\text{src}}}{d_r},$$

*and*

$$\mathbb{E}[a_i a_j] = \delta_{ij} \frac{A_{q,r}}{\kappa_i}.$$

*Proof.* The Euler–Lagrange equation for entropy with normalization and quadratic-energy constraints gives a Gaussian density with inverse temperature  $\beta_r$ . The expected energy is

$$E_{q,r}^{\text{src}} = \frac{1}{2} \sum_i \kappa_i \frac{1}{\beta_r \kappa_i} = \frac{d_r}{2\beta_r}.$$

Thus  $A_{q,r} = \beta_r^{-1} = 2E_{q,r}^{\text{src}}/d_r$ . Strict concavity of entropy gives uniqueness.  $\square$

*Remark 3.6* (discrete quotient branch). For a discrete finite quotient with intrinsic base weight  $m_r(q)$ , the exact Gibbs law is

$$\mu_r(q) = Z_r^{-1} m_r(q) e^{-\beta_r E_r(q)}.$$

A Gaussian statement on that branch requires a separate Laplace or central-limit theorem with controlled higher-order terms.

**Definition 3.7** (source release energy). *Let  $m_r^0$  be the tracially pointed base weight on the finite physical collar quotient. Let  $F_{r,a}$  be a finite ledger of primitive collar observables: scalar occupancy, protected-center presence, released volume, conserved source charges, and the release clock. The ledger excludes screen energy, sky spectra, CMB residuals, likelihoods, fitted amplitudes, and measurement-calibrated proxies. For source-emitted constraint values  $c_{r,a}^{\text{col}}$ , the selected release law is*

$$\nu_r^{\text{rel}}(x) = \frac{m_r^0(x) \exp[-\sum_a \lambda_{r,a} F_{r,a}(x)]}{Z_r}, \quad \mathbb{E}_{\nu_r^{\text{rel}}} F_{r,a} = c_{r,a}^{\text{col}}.$$

*Strict concavity of relative entropy gives uniqueness on full support. A boundary optimum uses the same form on its certified support. The constraint ledger, base weight, source DAG, multipliers, support, and refinement maps belong to the receipt.*

*For this independently selected quotient ensemble on released collar normal forms, define*

$$E_{q,r}^{\text{src}} = \frac{1}{2} \int \langle q_r(x), K_r q_r(x) \rangle_r d\nu_r^{\text{rel}}(x).$$

*The receipt must expose  $\nu_r^{\text{rel}}$ , the base measure,  $K_r$ ,  $d_r$ ,  $E_{q,r}^{\text{src}}$ ,  $A_{q,r}$ , no-observation ancestry, and the same operator normalization used in the action. MaxEnt alone does not determine amplitude.*

**Proposition 3.8** (microscopic-law necessity). *For fixed positive  $K_r$ , the Gaussian family  $Z(A)^{-1} \exp[-\langle q, K_r q \rangle / (2A)]$  exists for every  $A > 0$ . The quadratic MaxEnt form therefore leaves one positive scale free. The source release law in Definition 3.7, or an equivalent source-only vacuum law, supplies the energy that fixes  $A_{q,r} = 2E_{q,r}^{\text{src}}/d_r$ .*

**Theorem 3.9** (rotational screen spectrum). *If the continuum precision  $K_\theta$  commutes with the  $SO(3)$  action on scalar functions, then by Schur’s lemma each  $\mathcal{H}_\ell$  is an eigenspace. For the exact conformal-shell family,*

$$\Lambda_\ell(\theta) = \frac{\Gamma(\ell + 2 + \theta/2)}{\Gamma(\ell - \theta/2)}, \quad \ell \geq 2.$$

*Then  $\Lambda_\ell(0) = \ell(\ell + 1)$ , and the MaxEnt covariance gives*

$$C_\ell^q = A_q \frac{\Gamma(\ell - \theta/2)}{\Gamma(\ell + 2 + \theta/2)}.$$

*Consequently  $\mathcal{D}_\ell^q = \ell(\ell + 1)C_\ell^q / (2\pi) \sim (A_q/2\pi)\ell^{-\theta}$ .*

**Definition 3.10** (source-derived conformal precision). *Let  $L_r \succeq 0$  be the normalized detailed-balanced scalar collar Dirichlet operator on  $V_r$ , with continuum limit  $-\Delta_{S^2}$ . Set*

$$B_r = (L_r + \tfrac{1}{4}I)^{1/2}, \quad K_{\theta,r} = \frac{\Gamma(B_r + \frac{3}{2} + \frac{\theta}{2})}{\Gamma(B_r - \frac{1}{2} - \frac{\theta}{2})}.$$

*For  $-2 < \theta < 4$  on the retained  $\ell \geq 2$  branch, this operator is positive. The gamma recurrence gives  $K_{0,r} = L_r$  exactly. The finite receipt carries detailed balance, positivity, the  $\ell(\ell+1)$  normalization, anisotropy splitting, and refinement residuals.*

**Theorem 3.11** (finite refinement error). *Suppose that for  $2 \leq \ell \leq L$*

$$\left| \frac{\lambda_{\ell m,r}}{\Lambda_\ell(\theta)} - 1 \right| \leq \varepsilon_{K,r}(L), \quad \left| \frac{A_{q,r}}{A_q} - 1 \right| \leq \varepsilon_{A,r}, \quad \varepsilon_{K,r} < 1.$$

*Then  $C_{\ell m,r}^q = A_{q,r}/\lambda_{\ell m,r}$  satisfies*

$$\left| \frac{C_{\ell m,r}^q}{A_q/\Lambda_\ell(\theta)} - 1 \right| \leq \frac{\varepsilon_{A,r} + \varepsilon_{K,r}}{1 - \varepsilon_{K,r}}.$$

*Proof.* Write  $A_{q,r} = A_q(1 + a_r)$  and  $\lambda_{\ell m,r} = \Lambda_\ell(1 + b_{\ell m,r})$ . Then

$$\frac{C_{\ell m,r}^q}{A_q/\Lambda_\ell} = \frac{1 + a_r}{1 + b_{\ell m,r}},$$

and the displayed bound follows from  $|a_r| \leq \varepsilon_A$ ,  $|b_{\ell m,r}| \leq \varepsilon_K$ .  $\square$

**Theorem 3.12** (refinement-semigroup tilt). *Let  $R_b$  be the scalar refinement map for scale ratio  $b > 1$ . If one isolated scalar covariance mode has positive eigenvalue  $\lambda(b)$ ,  $\lambda(b_1 b_2) = \lambda(b_1)\lambda(b_2)$ , and  $\lambda$  is continuous, then there is a unique real  $\theta$  with*

$$\lambda(b) = b^{-\theta}, \quad \theta = -\frac{\log \lambda(b)}{\log b}.$$

*Proof.* Set  $g(t) = -\log \lambda(e^t)$ . The semigroup law gives  $g(t+s) = g(t) + g(s)$ . Continuity makes  $g(t) = \theta t$ , hence the result.  $\square$

**Definition 3.13** (edge-center reserve generator receipt). *Write  $s = \log b$  for logarithmic refinement thickness. Let  $u_{\text{full}}(s)$  be the scalar-conditioned covariance-survival cocycle across a full oriented collar, and let  $u_q(s)$  be its source-facing half-collar restriction. The receipt requires*

$$u(s+t) = u(s)u(t), \quad u(0) = 1,$$

*strong continuity at zero, the source-derived full-collar density*

$$-u'_{\text{full}}(0) = \frac{P_\star}{24},$$

*and the orientation-reversal coarea identity*

$$-u'_q(0) = \frac{1}{2}[-u'_{\text{full}}(0)] = \frac{P_\star}{48}.$$

*These are infinitesimal generator statements. A one-step survival probability has exponent  $-\log u_q(\log b)/\log b$ .*

**Theorem 3.14** (edge-center tilt and repair-clock reconciliation). *Under Definition 3.13,*

$$u_q(s) = e^{-\theta s}, \quad \theta = \frac{P_\star}{48}, \quad n_s = 1 - \frac{P_\star}{48}.$$

*In the coordinate  $\theta = \kappa_{\text{rep}}(P_\star - \varphi)$ , the same branch has*

$$\kappa_{\text{rep}}^{\text{edge}} = \frac{P_\star}{48(P_\star - \varphi)}.$$

*The value  $e$  is a separate diagnostic hypothesis unless an additional identity equates it with this source-derived coordinate.*

*Proof.* For  $g(s) = -\log u_q(s)$ , the cocycle law gives the continuous Cauchy equation. Hence  $g(s) = \theta s$ . Differentiation at zero and the half-collar identity give  $\theta = P_\star/48$ . The formula for  $\kappa_{\text{rep}}^{\text{edge}}$  is algebraic.  $\square$

**Theorem 3.15** (homogeneous radial source family). *Assume source-stress closure, a single clock, freezeout, multicenter consistency, and translation/rotation covariance. Let  $C_\zeta = M_{\Delta_\zeta^2}$  on the common physical  $d \ln k$  mode basis. If a scale-natural source embedding transports refinement to physical dilation and its finite operator residual converges to*

$$D_s^{-1} C_\zeta D_s = e^{-\theta s} C_\zeta, \quad (D_s f)(k, \hat{k}) = f(e^{-s} k, \hat{k}),$$

*then every positive measurable representative satisfies*

$$\Delta_\zeta^2(k) = A_\zeta (k/k_\star)^{-\theta}$$

*almost everywhere. Continuity gives the pointwise identity. A single shell has an infinite-dimensional radial kernel, including positive ambiguities; complete radial cross-covariances give the independent spherical-Hankel tomography route.*

*Proof.* Conjugation of the multiplication operator gives  $\Delta_\zeta^2(e^s k) = e^{-\theta s} \Delta_\zeta^2(k)$  almost everywhere. In logarithmic coordinates, rational-translation invariance on a common full-measure set forces  $e^{\theta t} \Delta_\zeta^2(e^t)$  to be constant almost everywhere. The one-shell and tomography statements follow from the correlation-restriction and spherical-Hankel theorems in the primordial bridge packet.  $\square$

**Theorem 3.16** (thin-shell power-law lift). *Assume  $q(\hat{n}) = Z_\star \Pi_{\ell \geq 2} \zeta_\star(R_\star \hat{n})$  and*

$$\Delta_\zeta^2(k) = A_\zeta (k/k_\star)^{-\theta}.$$

*Then*

$$C_\ell^q = 4\pi Z_\star^2 \int d \ln k \Delta_\zeta^2(k) j_\ell^2(k R_\star),$$

*and, for  $-2 < \theta < 4$ ,*

$$A_q = \pi^{3/2} Z_\star^2 A_\zeta (k_\star R_\star)^\theta \frac{\Gamma(1 + \theta/2)}{\Gamma(3/2 + \theta/2)}.$$

*Thus*

$$A_\zeta = \frac{A_q}{\pi^{3/2} Z_\star^2 (k_\star R_\star)^\theta} \frac{\Gamma(3/2 + \theta/2)}{\Gamma(1 + \theta/2)}.$$

**Proposition 3.17** (finite-window bound). *Let  $\Psi_\ell(k) = \int dr W(r) j_\ell(kr)$ ,  $W \geq 0$ ,  $\int W dr = 1$ , with mean  $R_\star$  and variance  $\sigma_R^2$ . If*

$$\delta_\ell(k) = \frac{k^2 \sigma_R^2}{2} \sup_{r \in \text{supp } W} |j_\ell''(kr)|,$$

*then*

$$|C_{\ell,W}^q - C_{\ell,\text{shell}}^q| \leq 4\pi Z_\star^2 \int d \ln k \Delta_\zeta^2(k) \left[ 2\delta_\ell(k) + \delta_\ell(k)^2 \right].$$

**Proposition 3.18** (radial non-identifiability). *For a finite radial basis,  $C = Ap$ . If  $A \in \mathbb{R}^{N_\ell \times N_k}$  has rank  $r$ , then  $\dim \ker A = N_k - r$ , and every  $p+v$  with  $v \in \ker A$  gives the same screen spectrum. A physical radial reconstruction therefore requires either a source theorem restricting  $p$ , or a declared prior with a published null basis, resolution kernels, positivity checks, and prior-sensitivity report.*

**Theorem 3.19** (conditional source spectrum). *One hash-locked source DAG may emit*

$$(q_r, S_{\text{scr},r}, \theta, A_{q,r}, C_{\ell,r}^q, A_{\zeta,r}, \Delta_{\zeta,r}^2)$$

*as a conditional primordial packet only when the geometric-scalar, collar-law, release-energy, reserve-generator, conformal-precision, refinement, source-stress, single-clock, freezeout, adiabaticity, isocurvature, phase-coherence, thin-shell, radial-null, finite-window, and forward- residual receipts pass on that DAG. Any measurement, likelihood, fitted parameter, or measurement-calibrated ancestor fails the source packet. Physical temperature and polarization spectra require the independent Boltzmann, recombination, nuisance, covariance, and likelihood gates.*

**Target 3.20** (requirements for a source-derived screen spectrum). *A concrete source-derived spectrum requires the following receipt families before primordial interpretation:*

1. *geometry, scalar quotient, low-mode projector, scalar precision, and operator normalization;*
2. *quotient ensemble selection, scalar release energy, and MaxEnt screen covariance;*
3. *the infinitesimal reserve generator, scalar refinement tilt, and angular screen spectrum with a finite error budget;*
4. *source-stress, clock, freezeout, radial-window, radial-null, forward-residual, and transfer fire-wall receipts.*

*The theorem and algorithm packet is complete. Construction of one finite source DAG that passes this receipt set is work in progress.*

## 4 Branch Thesis

The branch claim is:

On a finite-source branch, OPH may replace the logical jobs of inflation by clock-slice spatial-holonomy selection, same-boundary or low- $k$  repair coherence, a geometric screen spectrum, MaxEnt release, and ordinary Boltzmann transfer.

This claim lies outside the recovered Standard Model and general-relativistic core. It becomes predictive only when the source objects are fixed before observational comparison. Screen synchronization implements modular self-consistency, with internal time generated by the state-algebra pair rather than by an external clock.

Coherent-matter same-channel forcing supplies a local scalar-channel identity for nondegenerate OPH-coherent material sources under Scalar Edge-Center Exhaustion. It does not supply the finite covariant dark-stress parent needed for cosmological  $w_A$ ,  $c_{s,A}^2$ ,  $\sigma_A$ ,  $Q_A^\mu$ , recipient-stress, gauge-variable, cold-dark-matter (CDM) limit, and refinement-convergence receipts.

The repair-charge dark-sector construction supplies a source-only  $\bar{\rho}_A(a)$  only when a finite source fixes the anomaly abundance. Without that source, the dark continuation is unspecified.

## 5 Physical Conditions

| Physical question     | Required result or premise                                                                                                                                                                                                            |
|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Finite early state    | Quotient normal-form branch with declared base measure and implementation invariance.                                                                                                                                                 |
| Flatness              | Zero clock-slice spatial Levi–Civita holonomy identifies $\kappa = 0$ . Exact selection requires a derivation from the finite screen or an explicit curvature premise. Damping gives only an approximate $ K $ or $ \Omega_K $ bound. |
| Horizon coherence     | Same-boundary unique-extension theorem or a uniform low- $k$ repair gap.                                                                                                                                                              |
| Near scale invariance | Geometric screen scalar, normalized scalar precision, source release energy, infinitesimal reserve-generator tilt, and angular screen-spectrum receipts.                                                                              |
| Amplitude             | The conditional theorem fixes $A_q = 2E_q^{\text{src}}/d$ and its source-family conversion to $A_\zeta$ . A finite collar release law and energy receipt must supply their values without CMB data.                                   |
| Hot start             | MaxEnt release into the realized radiation branch with a common release clock.                                                                                                                                                        |
| Acoustic transfer     | Declared source functions and a Boltzmann transfer calculation.                                                                                                                                                                       |

## 6 Numerical Acceptance Conditions

The simulator reference is <https://github.com/muellerberndt/oph-physics-sim>. For this paper the simulator may emit diagnostic screen spectra, observer-screen synchronization receipts, and release-amplitude candidates. It supports an inflation-replacement interpretation only when one finite source supplies the primordial growth mode, physical radial lift, transfer calculation, and likelihood inputs.

## 7 Flatness Boundary

The flatness branch is split into identification and selection. On a clocked Friedmann–Lemaître–Robertson–Walker (FLRW) continuation with  $u_a = -N\nabla_a\chi$ ,  $h_{ab} = g_{ab} + u_a u_b$ , and  $K(\tau) = \kappa/a(\tau)^2$ , the area-normalized small-loop holonomy of the spatial Levi–Civita connection identifies  $K = 0$  and hence  $\kappa = 0$ . It does not select that sector from the raw FLRW fiber. A numerical calculation must distinguish exact selection derived from the finite screen, a flat branch conditioned on an explicit

premise, and an open selection problem. Screen and collar defect decay is a repair diagnostic rather than a spatial curvature holonomy. Curvature damping supplies only an approximate bound on  $|K|$  or  $|\Omega_K|$ .

## 8 Screen-Spectrum Branch

The load-bearing screen result is the angular screen covariance. It is distinct from a CMB spectrum:

$$C_\ell^q = A_q \frac{\Gamma(\ell - \theta/2)}{\Gamma(\ell + 2 + \theta/2)}.$$

This statement requires the geometric  $q_r$  of Definition 3.2, a normalized conformal precision operator  $K_{\theta,r}$ , the primitive collar release law,  $E_{q,r}^{\text{src}}$ , the MaxEnt covariance theorem, and a finite-to-continuum refinement error budget. The edge-center theorem gives

$$\theta = \frac{P_\star}{48}, \quad n_s = 1 - \frac{P_\star}{48}, \quad \kappa_{\text{rep}}^{\text{edge}} = \frac{P_\star}{48(P_\star - \varphi)}.$$

Its premise is a source-emitted full-collar generator density  $P_\star/24$  with the orientation-half identity. A finite one-step survival probability is converted through  $-\log u/\log b$ . The  $e(P_\star - \varphi)$  clock branch is a diagnostic alternative. The finite generator and clock receipt is work in progress.

Under the declared source dilation cocycle, the primordial power law

$$\Delta_\zeta^2(k) = A_\zeta \left( \frac{k}{k_\star} \right)^{-\theta}$$

is the unique member of the one-dimensional source family. The exact thin-shell identity fixes the amplitude conversion

$$A_\zeta = \frac{A_q}{\pi^{3/2}} \frac{\Gamma(3/2 + \theta/2)}{\Gamma(1 + \theta/2)}$$

at the source pivot. A finite run requires  $R_\star$ , the radial window, the unrestricted null-space report, the forward residual, and the physical scale. The screen theorem proves no temperature, polarization, lensing, acoustic-peak, or likelihood statement.

## 9 Open Premises

The same-boundary or low- $k$  coherence branch has no quotient-language derivation. A finite source emitting the geometric collar-volume field

$$q_r = \Pi_{\geq 2,r}(1/3) \log(J_{X,r}/\bar{J}_{X,r})$$

is work in progress. The same source must supply the primitive collar release law and pooled release energy that fix  $A_q$ , together with the full-collar infinitesimal reserve generator and orientation-half condition used by the conditional  $P_\star/48$  theorem. Its source-dependency graph must contain compatible stress, clock, freezeout, physical-scale, radial-window, null-space, and forward-residual evidence. The resulting quantities are the inputs to the cosmic microwave background transfer and likelihood calculations.